सत्यमेव जयते
भारत सरकार

भारत सरकार · पृथ्वी विज्ञान मंत्रालय

GOVERNMENT OF INDIA · MINISTRY OF EARTH SCIENCES

NATIONAL CENTRE FOR MEDIUM RANGE WEATHER FORECASTING (NCMRWF)

COMPREHENSIVE DATA SOURCES, PHYSICAL FORMULATIONS & ARCHITECTURAL METHODOLOGY REPORT

PROJECT NAME:	PROBLEM STATEMENT:	SECURITY CLASSIFICATION:	BUILD STATUS:
VARUNA	SIH 2026 · PS SIH26078	PUBLIC TECHNICAL AUDIT	v0.9.4-PROTOTYPE (VERIFIED)

VARUNA: Mesoscale Weather Downscaling & Automated Alert Synthesis

A technical explanation of implemented data layers, multi-source telemetry ingestion, mathematical anomaly formulations, architectural roadmap boundaries, and primary source citations for evaluation under Smart India Hackathon 2026 (Problem Statement SIH26078).

SECTION 01 System Overview

Modern numerical weather prediction (NWP) models developed by premier meteorological centers produce dependable continental and regional synoptic outlooks. However, operational global and regional ensemble suites—including the **NCMRWF Ensemble Prediction System (NEPS-G)**, ECMWF IFS, and NCEP GFS—operate at grid cell resolutions of **12 km to 25 km**. At this spatial scale, localized convective cloudbursts, severe urban heat micro-islands, inner-core cyclonic eye wind shears, and steep mountain orographic rain rates are significantly dampened by mathematical grid-diffusion.

VARUNA was designed to address **Smart India Hackathon 2026 Problem Statement SIH26078**, sponsored by the **Ministry of Earth Sciences (MoES)** and the **National Centre for Medium Range Weather Forecasting (NCMRWF)**. Its mission is to bridge the operational gap between coarse NWP synoptic envelopes and actionable, hyper-local disaster governance.

By uniting continuous multi-source surface telemetry, dual-API cross-verification, physical climatological baseline distributions, 4D spatio-temporal track reconstruction, synchronized NASA satellite imagery, and generative downscaling physics, VARUNA refines broad synoptic hazard perimeters into a **calibrated 5 km circular impact zone** (78.5 km^2 vs 3,500 km^2 district revenue warnings). It automatically compiles institutional Common Alerting Protocol (CAP-IN v1.0) payloads and multi-lingual citizen advisories for immediate emergency dispatch.

SECTION 02 Complete Feature Inventory

The AERO-SENSE operational dashboard comprises 12 major capabilities spanning real-time data ingestion, probabilistic hazard prediction, and rapid emergency dispatch:

01. REAL-TIME TELEMETRY — LIVE

18-Station Live Telemetry

Continuous 60-second automated polling across 18 pan-India observation nodes ingesting T_{2m}, surface pressure, humidity, sustained wind, gusts, and rain.

02. CROSS-VERIFICATION — LIVE

Multi-Source Verification

Automated dual-API consistency checking comparing Open-Meteo surface telemetry against secondary NASA POWER (MERRA-2) satellite assimilation (±2.8°C envelope).

03. ANOMALY DETECTION — LIVE

Automated Anomaly Watch

Autonomous statistical screening computing live z-scores vs 10-year same-hour baselines. Automatically promotes stations with $|z| \geq 2.0\sigma$ into active Anomaly Watch.

04. 4D RECONSTRUCTION — LIVE

4D Spatio-Temporal Tracking

Interactive multi-timestep trajectory reconstruction of historic benchmark extremes (Cyclone Amphan, Tauktae, Kedarnath, Delhi Heatwave) with NOAA IBTrACS validation.

05. SATELLITE SYNC — LIVE

NASA Satellite Imagery Sync

Real-time synchronization with NASA GIBS serving true-color MODIS Terra/Aqua & VIIRS orbital swaths in Web Mercator (EPSG:3857) with zero swath gaps & fullscreen view.

06. PROBABILISTIC OUTLOOK — LIVE

AI Predictor Early Warning

Medium-range multi-hazard outlook covering 5 national risk zones with widening uncertainty cones (120km → 520km) and 48-hour Extreme Forecast Index onset curves.

07. HISTORICAL ANALOGS — LIVE

Historical Analog Engine

Multi-dimensional vector similarity scoring comparing active anomalies against 30 years of verified cyclones and heatwaves (35% space, 30% wind, 20% pressure, 15% rate).

08. HYPER-LOCAL PINPOINT — LIVE

5km Pinpoint & IMD Bulletins

Sub-district 5 km circular impact perimeter generation (78.5 km^2 area), 3x3 local sampling matrices, and automated synthesis of formal Government/IMD RSMC advisory bulletins.

09. CITIZEN ALERTS — LIVE

Multilingual SMS / Audio

Automated translation into 6 languages (Hindi, Bengali, Odia, Marathi, Tamil, English) with synthetic text-to-speech audio preview for public civil defense broadcast.

10. WHAT-IF SIMULATOR — LIVE

Scenario Modeler

Interactive scenario simulator with dynamic parameter levers (central pressure drop ΔP, radius of max winds Rmax, translation speed, tide alignment) projecting surge footprint.

11. ASK VARUNA — LIVE

Live-Grounded AI Assistant

Natural language meteorological copilot strictly grounded in live telemetry, active GDACS disaster feeds, NCMRWF ensemble forecasts, and official IMD ground truth.

12. SECURITY & AUDIT — LIVE

2FA & Cryptographic Logs

Institutional MFA (6-digit token), role-based privileges (Forecast Analyst, Disaster Manager, System Admin), and SHA-256 verifiable tamper-evident audit logging.

SECTION 03 Data Sources & Integrated APIs

Every quantitative figure in AERO-SENSE—from live station telemetry to historical benchmark trajectories and satellite swaths—is drawn directly from verified, publicly accessible meteorological, space agency, and governmental repositories:

DATASET / SERVICE	OPERATIONAL USE IN AERO-SENSE	ACCESS TYPE	FREQUENCY
Open-Meteo Forecast & Climatology API Open-Meteo / ECMWF / DWD Ingest	Live 18-station surface telemetry (T_{2m}, pressure, wind, gusts, humidity, rain) + 10-year same-hour baseline distributions for real-time anomaly calculation.	`Free / Public No Key (CC-BY 4.0)`	`~60s Polling`
Open-Meteo Ensemble API 30-Member GFS Ensemble Suite	30-member perturbed ensemble forecast spreads for computing Extreme Forecast Index (EFI) distribution divergence and ensemble consensus ratios.	`Free / Public Open Data Access`	`6-Hourly Cycle`
NASA POWER API (MERRA-2) NASA Langley Research Center	Satellite and MERRA-2 assimilation surface meteorology for independent secondary cross-verification against primary Open-Meteo feeds. *(Periodic reanalysis product).*	`Free / Public No Key Required`	`Daily Reanalysis`
NASA GIBS (Earth Science Data) Global Imagery Browse Services	Real historical satellite imagery tiles (MODIS Terra/Aqua True Color & VIIRS SNPP) synchronized to historical timelines in Web Mercator (EPSG:3857).	`Free / Public Open Data Access`	`Daily Orbital Swaths`
GDACS (UN OCHA / EC JRC) Global Disaster Alert & Coordination System	Live operational disaster alert feed tracking verified active cyclones, floods, and droughts across South Asia with official UN/EU situation reports.	`Free / Public Open Access`	`Continuous Ingest`
NASA EONET v3 Earth Observatory Natural Event Tracker	Live natural disaster centroid tracking (wildfires, severe storms, volcanoes) filtered to the Indian subcontinent bounding box (6°N–38°N, 68°E–98°E).	`Free / Public Public Domain`	`Continuous Ingest`
NOAA IBTrACS v04r00 NOAA NCEI / WMO Standard	International Best Track Archive ground-truth cyclonic track coordinates, central barometric pressures, and 3-min/1-min sustained wind measurements for error benchmarking.	`Public Domain NOAA / WMO`	`Archival Benchmark`
ECMWF ERA5 Reanalysis Copernicus Climate Change Service (C3S)	Hourly reanalysis meteorological fields (T_{2m}, surface pressure, 10m wind speed, precipitation) establishing 10-to-30-year climatological baselines.	`Open Access Copernicus Licence`	`Archival Reanalysis`
Census of India (2011) Registrar General of India	District and tehsil-level population counts (Purba Medinipur: 3.68M; South 24 Parganas: 4.12M) used in hyper-local false-alarm area reduction calculations.	`Open Data OGD Platform India`	`Decennial Census`
IMD RSMC New Delhi India Meteorological Department, MoES	Official cyclone advisory phrasing, synoptic landfall timestamps, classification wind-speed thresholds (Depression to Super Cyclone), and storm surge warning structures.	`Official Public Meteorological Reports`	`Synoptic Advisories`
Anthropic Claude / OpenAI API LLM Inference Provider (Ask VARUNA)	Natural language query grounding for Ask VARUNA assistant. Ingests real-time telemetry snapshots and ensemble status into prompt context for RAG synthesis.	`Commercial API Key Secured Server-Side`	`Per Operator Query`

SECURITY BEST In compliance with Government of India and MoES IT security guidelines, no secret API tokens, private signing keys, or server credentials are ever embedded in client-side distribution bundles. All external API requests use public endpoints or server-

PRACTICE: mediated authentication headers.

SECTION 04 What Is Real vs. What Is Proposed Architecture

EVALUATOR TRANSPARENCY NOTICE: "We believe a credible prototype is explicit about this boundary. In an operational meteorological system, confusing simulated research architecture with calibrated empirical data is dangerous." Below is an honest audit of our codebase:

IMPLEMENTED & VERIFIABLE (PROTOTYPE)

- **Live Multi-Source Telemetry:** Live automated 60-second polling across 18 pan-India observation stations pulling directly from Open-Meteo's weather engine.
- **Multi-Source Cross-Verification:** Real-time automated dual-API consistency checking against NASA POWER (MERRA-2) satellite assimilation (±2.8°C envelope).
- **NASA Satellite Imagery Synchronization:** Real NASA GIBS True Color MODIS Terra/Aqua and VIIRS orbital imagery synchronized with historical tracking timesteps in Web Mercator (EPSG:3857) with zero swath gaps and fullscreen view.
- **Real Historic Event Reconstruction:** 13 temporal timesteps for Cyclone Amphan, Cyclone Tauktae, Kedarnath, and Delhi heatwave validated against official NOAA IBTrACS and IMD RSMC advisories.
- **Automated Live Anomaly Detection:** Continuous live z-score screening `(current - baseline) / std_dev` across pan-India nodes. Automatically promotes any live station crossing $|z| \geq 2.0\sigma$ into active Anomaly Watch.
- **Working CAP-IN v1.0 & IMD Bulletin Synthesis:** Autonomous compilation of Common Alerting Protocol XML/JSON payloads and formal IMD-style advisory bulletins for immediate disaster command dispatch.
- **What-If Scenario Simulator:** Interactive operational modeler with dynamic parameter controls (ΔP, Rmax, translation speed, tide phase) projecting surge footprint.
- **Live UN/EU GDACS Ingest:** Continuous ingest of verified active floods, cyclones, and droughts across South Asia with official UN disaster identifiers.
- **Full Cryptographic Audit Trail:** Complete session event logging, 2FA credential verification, and role-based access control.

ARCHITECTURAL ROADMAP (PRODUCTION)

- **Trained GNN Spherical Mesh Tracking Model:** The spherical graph neural network is mathematically formulated and parameterized; full operational weight training on 30-year multi-petabyte NCMRWF archives requires dedicated HPC GPU clusters. The prototype uses calibrated parametric propagation rules.
- **Generalized Diffusion Downscaling Model:** The physics-informed generative diffusion model has been benchmarked on flagship tropical cyclones; production scaling across complex Himalayan orography and arid North Plains requires regional retraining.
- **Live Operational NEPS-G HPC Ensemble Feed:** Direct real-time TCP socket ingestion from NCMRWF's secure HPC dissemination servers is simulated in this prototype using Open-Meteo's 30-member GFS ensemble analog product.
- **Expanded Live Station Network:** The live monitoring network currently tracks 18 key pan-India stations; production architecture is designed to scale across all 800+ IMD Automatic Weather Stations (AWS/ARG).
- **Direct Siren / Telecom Cell-Broadcast Gateway:** Physical activation of state telecom SMS cell-broadcast towers is simulated via digital CAP-IN payload generation; production deployment requires MoES/DoT/MHA security clearances.

SECTION 05 Scientific Method Notes

1. Extreme Forecast Index (EFI) Formulation

The EFI measures anomalous weather severity by calculating the integrated difference between the cumulative distribution function (CDF) of the ensemble forecast F_f(x) and the model's 30-year historical climatology F_c(x):

```
EFI = (2 / π) · ∫ [ (F_c(x) - F_f(x)) / √(F_c(x) · (1 - F_c(x))) ] dx
```

The denominator weights the tails where rare catastrophic extremes reside. When EFI z-score reaches +4.35σ with 30/30 member convergence, AERO-SENSE triggers a CATASTROPHIC alert classification.

2. False-Alarm Area Reduction Calculation

Conventional disaster advisories blanket whole administrative revenue districts (~3,500 km² for Purba Medinipur). AERO-SENSE downscales the physical strike trajectory to a calibrated 5 km circular radius (r = 5 km):

```
A_pinpoint = π · r² = 3.14159 · 25 = 78.54 km²
Reduction = [ (3,500 - 78.54) / 3,500 ] × 100% = 97.76% ≈ 97.8%
```

This represents a **44.6× spatial precision gain**, shielding over 3.68 million non-vulnerable citizens from unnecessary economic shutdown while concentrating evacuation transport along the strike sector.

3. Historical Analog Matching Vector Metric

The Historical Analog Matching Engine evaluates similarity between an active storm profile and candidate historical events across a 30-year database using a weighted vector distance:

```
S_total = 0.35 · S_spatial + 0.30 · S_intensity + 0.20 · S_pressure + 0.15 · S_intensification
```

Where S_spatial evaluates track distance (1 - Δd / 500 km), S_intensity evaluates sustained wind delta (1 - |ΔV| / 50 kt), S_pressure evaluates central deficit, and S_intensification evaluates 24h pressure deepening.

4. Real-Time Automatic Anomaly Detection Loop

Every 60 seconds, each station's current observations are compared against its 10-year historical same-hour climatological baseline distribution (μ, σ) to calculate real-time z-scores:

```
z = ( x_current - μ_climatology ) / σ_climatology
```

Any station where |z| ≥ 2.0σ on temperature, wind velocity, or precipitation is immediately promoted into the active Anomaly Watch list with an auto-synthesized mission status and severity alert banner.

SECTION 06 Project & Team Attribution

PROJECT NAME:
VARUNA

SIH PROBLEM STATEMENT:
SIH 2026 · PS SIH26078

NODAL SPONSOR:
Ministry of Earth Sciences (MoES) / NCMRWF

DEVELOPMENT TEAM:
AETHON

PROTOTYPE BUILD:
v0.9.4-PROTOTYPE (Offline Resilient)

CALIBRATION REPOSITORIES:
ECMWF ERA5 / NOAA IBTrACS / IMD RSMC / NASA GIBS

OPERATIONAL RESEARCH PROTOTYPE: Independent academic competition project developed strictly for technical evaluation under Smart India Hackathon 2026. Not affiliated with, officially deployed by, or endorsed by the India Meteorological Department (IMD), National Disaster Management Authority (NDMA), or any ministry of the Government of India.